EXP 10: PERFORMANCE OPTIMIZATION IN ZOHO CREATOR

AIM

To study and apply performance optimization techniques in Zoho Creator so that the Library Book Issue Management application loads faster, executes scripts efficiently, handles a growing number of records and uses fewer resources.

PROCEDURE

1. **Identify the bottlenecks**
   - Open the application and check the Deluge execution logs for slow or failing workflows and functions.
   - Add *info* statements inside scripts to find which part takes the most time.
   - Note which forms, reports and pages take the longest to load, so that they can be improved first.
2. **Optimize the data model (forms)**
   - Keep only the required fields in each form; remove unused and duplicate fields.
   - Use lookup fields to link forms (for example, Book_Catalog) instead of copying the same data again.
   - Avoid too many subforms and heavy formula fields in a single form.
3. **Optimize Deluge scripts**
   - Fetch records with specific criteria and a limited range instead of fetching all records.
   - Do not fetch or update records inside a loop; fetch once and then process the collection.
   - Use aggregate functions such as count and sum instead of looping through all the records.
   - Use try-catch blocks so that an error does not stop or slow the whole workflow.

```
// Slow: fetches every record and counts them in a loop
all = Book_Issue_Request[ID != 0];
total = 0;
for each r in all { if(r.Issue_Status == "Requested") { total = total + 1; } }

// Optimized: criteria + aggregate function
total = Book_Issue_Request[Issue_Status == "Requested"].count();
```

4. **Use bulk operations**
   - Update many records in a single statement instead of updating them one by one in a loop.
   - This reduces the number of database calls and the script execution time.

```
// Bulk update with criteria (no loop)
Book_Issue_Request[Issue_Status == "Requested"].Issue_Status = "Approved";
```

5. **Optimize workflows**
   - Add conditions so that a workflow runs only when it is needed, and choose the correct form event.
   - Move heavy tasks such as fine calculation and reminders to scheduled or batch workflows that run at off-peak time (for example, 09:00:00 daily).
   - Disable workflows that are no longer used.
6. **Optimize reports**
   - Show only the necessary columns and apply default filters (Book Category, Member Type, Issue Status).
   - Use pagination so that a limited number of records is loaded at a time.
7. **Optimize pages and dashboards**
   - Limit the number of embedded reports, charts and HTML snippets on one page.
   - Prefer summary values such as counts and gauges instead of loading full lists on the dashboard.

8. **Housekeeping**
   - Archive or delete old, unused and test records regularly to keep the data size small.
9. **Test and compare**
   - Test the application with sample data after the changes and compare the loading and execution time with the earlier results.
10. **Monitor regularly**
   - Review the execution logs and the report and page load times at regular intervals, especially when the number of records increases.
   - Repeat the optimization steps whenever a new form, workflow or report is added to the application.

Summary of Techniques

Area	Technique	Benefit
Forms	Fewer fields, lookups, limited subforms	Faster form loading and saving
Deluge	Criteria, range, aggregate functions, no fetch in loops	Less execution time and fewer statements
Bulk operations	Update or insert many records at once	Fewer database calls
Workflows	Conditions, scheduled and batch workflows	Heavy jobs do not slow down users
Reports and pages	Filters, pagination, fewer components	Faster page and report loading
Data	Archive and delete old records	Better performance as data grows

Common Performance Problems and Solutions

Problem	Cause	Solution
Report takes long to open	Too many records and columns loaded at once	Add default filters, reduce columns and use pagination
Workflow runs slowly	Fetch or update statements placed inside loops	Fetch once with criteria and use bulk update
Form submission is slow	Many workflow actions running on every submit	Add conditions and move heavy tasks to scheduled or batch workflows
Dashboard loads slowly	Too many embedded reports and charts	Use counts, gauges and a limited number of components

Best Practices

- Always test changes in sample data before making them live.
- Keep scripts short, readable and commented so that they are easy to review and improve.
- Avoid repeating the same fetch statement; store the result in a variable and reuse it.
- Use filters and criteria wherever records are displayed or processed.
- Review the application regularly and remove components that are no longer needed.

RESULT

The performance optimization techniques for Zoho Creator were studied and applied successfully. By using an efficient data model, optimized Deluge scripts with criteria and bulk operations, conditional, scheduled and batch workflows, and lightweight reports and pages, the speed, scalability and reliability of the Library Book Issue Management application were improved. Regular monitoring and following the best practices keeps the application efficient as the number of users and records grows.